

LARAMIE, WY, USA

WMO: 725645

Lat: 41.317N

Lon: 105.683W

Elev: 7266

StdP: 11.23

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24022

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-9.9	-3.3	-14.9	3.2	-7.8	-9.0	4.4	0.0	39.8	33.8	35.4	32.3	5.8	250	0.779

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.2	85.3	55.0	82.9	54.5	80.7	54.0	59.3	74.8	58.0	73.7	56.9	72.6	13.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.8	84.0	62.1	53.2	79.0	61.5	51.5	74.2	60.6	29.2	74.5	28.1	73.3	27.2	71.9	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.5	28.9	26.3	-23.1	89.7	6.7	1.6	-27.9	90.9	-31.8	91.8	-35.6	92.7	-40.5	93.9		
(5)			-23.0	62.4	5.9	1.4	-27.2	63.4	-30.7	64.3	-34.0	65.1	-38.3	66.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	22.5	24.4	32.1	38.2	47.4	58.3	65.3	63.0	54.7	42.3	31.2	21.9
	DBStd	17.70	11.25	11.33	10.21	8.75	7.93	6.46	4.07	4.17	7.20	9.79	11.54	11.93
	HDD50	4376	852	717	555	361	142	13	0	0	37	263	564	872
	HDD65	8530	1317	1137	1019	805	546	212	45	85	311	703	1013	1337
	CDD50	1411	0	0	1	5	61	262	474	403	179	25	1	0
	CDD65	91	0	0	0	0	0	11	54	23	3	0	0	0
	CDH74	2762	0	0	0	0	29	519	1258	751	204	1	0	0
	CDH80	618	0	0	0	0	2	118	343	129	26	0	0	0
Wind	WSAvg	12.1	13.2	13.6	13.3	13.4	12.2	11.7	9.8	9.9	10.6	12.0	12.5	13.0
Precipitation	PrecAvg	10.8	0.4	0.4	0.7	1.0	1.5	1.4	1.4	1.1	1.0	0.8	0.6	0.4
	PrecMax	16.6	1.7	1.6	2.4	3.7	3.9	4.8	4.4	2.2	3.0	2.7	2.6	1.9
	PrecMin	5.4	0.0	0.0	0.0	0.2	0.0	0.1	0.2	0.4	0.0	0.0	0.0	0.0
	PrecStd	2.7	0.4	0.4	0.6	0.7	0.9	1.0	1.1	0.6	0.8	0.6	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	54.1	63.9	68.5	77.4	87.8	89.2	86.1	83.5	72.6	63.2	52.0
		MCWB	35.3	37.9	42.0	44.8	51.7	54.6	56.6	55.0	53.4	47.6	41.9	36.5
	2%	DB	45.4	48.5	58.6	64.5	73.0	83.5	86.4	83.3	79.2	68.6	57.7	45.8
		MCWB	33.8	35.1	39.8	43.0	49.6	53.0	55.9	54.6	52.0	46.2	39.8	34.1
	5%	DB	41.4	44.0	54.2	60.9	69.3	80.0	83.9	81.2	76.0	64.6	52.7	41.8
		MCWB	32.2	33.0	38.0	41.7	48.3	52.5	55.5	53.9	50.7	44.6	38.0	32.2
	10%	DB	37.8	40.1	49.1	56.0	64.9	76.4	81.4	78.8	72.6	60.0	47.9	38.2
		MCWB	30.3	31.5	36.0	40.2	46.8	51.9	55.0	53.7	49.8	43.0	35.8	30.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	38.9	43.2	47.3	55.4	58.7	62.0	60.3	57.4	50.2	43.4	37.3
		MCDB	47.1	51.1	60.2	64.9	71.0	74.4	77.4	76.7	73.2	65.2	58.0	48.6
	2%	WB	34.9	36.0	40.7	44.5	51.8	56.8	60.1	58.5	55.0	48.0	40.8	34.9
		MCDB	43.4	46.4	56.6	59.7	66.3	73.6	75.6	74.0	70.7	64.2	56.0	44.6
	5%	WB	32.8	33.7	38.6	42.6	50.0	55.2	58.8	57.2	53.4	46.0	38.6	32.8
		MCDB	40.9	42.6	52.3	57.6	65.0	72.4	74.3	72.4	68.7	61.4	51.6	41.1
	10%	WB	30.7	31.9	36.4	40.9	48.2	53.6	57.5	56.1	51.8	44.0	36.3	30.7
		MCDB	37.7	39.8	48.2	55.2	62.4	70.7	73.3	71.1	66.6	58.2	47.1	37.9
Mean Daily Temperature Range	5% DB	MDBR	20.3	20.9	23.2	24.1	25.5	29.7	30.2	29.7	29.3	25.7	22.2	20.2
		MCDBR	22.4	24.1	30.0	33.2	33.1	36.0	34.8	34.1	35.2	32.7	27.8	22.2
		MCWBR	14.3	14.5	15.3	16.1	14.2	13.4	11.8	11.9	14.2	15.6	14.9	13.7
	5% WB	MCDBR	21.0	21.7	27.9	30.0	29.6	31.1	28.8	28.1	28.9	29.3	26.4	21.0
		MCWBR	14.1	13.6	14.8	15.2	13.7	13.1	11.0	10.8	12.2	14.4	14.6	13.6
Clear-Sky Solar Irradiance	taub	0.220	0.230	0.255	0.277	0.297	0.301	0.329	0.326	0.285	0.254	0.234	0.225	
	taud	2.541	2.527	2.518	2.501	2.514	2.516	2.471	2.467	2.581	2.644	2.596	2.527	
	Ebn at Noon	310	320	320	317	310	308	298	296	305	307	302	298	
	Edh at Noon	22	26	30	32	33	33	34	33	28	23	20	20	
All-Sky Solar Radiation	RadAvg	738	1043	1432	1696	1912	2275	2147	1876	1598	1126	780	610	
	RadStd	29	52	115	113	161	119	85	64	103	84	57	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page